

Ali Syed Ayan

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Education

City University of Hong Kong

B.Sc. Computer Science, Minor in Business Intelligence

Hong Kong

Jun 2022 – Jun 2026

- Scholarships & awards: International Student Top Scholarship (HK\$720,000 total); Dean's List; Tiger Talent Program (3×).
- Coursework: Web Application Development, Big Data Analytics, Machine Learning, Data Structures & Algorithms, Database Systems, Fundamentals of Data Science, Operating Systems, Computer Networks, Software Engineering, Discrete Mathematics, Computer Vision, Data Science.

Work Experience

Hong Kong Monetary Authority

Analyst Programmer (IT Division – Business Enablement & Software Architecture)

Hong Kong

Jul 2025 – Present

AI / ML, data platforms, and automation

- Engineered an open-source chatbot for BRDR using retrieval-augmented generation (RAG) over a knowledge base of about 10,000 PDFs; Python for ingestion and processing, **pgvector** for embeddings, SQL-backed vector search, and Grafana for observability.
- Built an agentic firewall review automation tool using LLMOps practices (JIRA API, Next.js, Open Policy Agent) to encode review guidelines as policy-as-code and LLM-friendly workflows, reducing security review effort by about 30 hours/month.

Product delivery, operations, and stakeholder collaboration

- Spearheaded an AI Outlook Email Assistant within a Scrum team across the full SDLC; served as L3 support for production troubleshooting and incident resolution.
- Worked with the Product Owner and engineers to translate requirements into technical workflows; delivered training, prompts, and documentation; produced Tableau and Power BI dashboards for reporting.

Jay Cee Ess Global

Software Engineer

Remote (Australia-based)

Aug 2025 – Present

Growth, conversion, and product

- Participated in performance tracking that supported about AUD 100,000/month revenue; helped revamp a legacy site into a high-performance web app, run Google Ads, improve UX (about 200% user intake), and lift conversion rates.

Payments and e-commerce

- Reduced cart abandonment about 70% by integrating Stripe with Apple Pay, Google Pay, and Alipay for multi-method checkout.

Algorithms and performance engineering

- Cut price retrieval latency about 99% with a custom in-memory routing engine based on Yen's *K*-shortest paths algorithm.

Data layer and infrastructure

- Reduced database costs about 75% by replacing a rigid Supabase setup with Vercel Upstash Redis for fast lookups and complex multi-carrier routing, unlocking new logistics market segments.

Universal Innotech Group Company Limited

Junior Software Engineer

Tsuen Wan, Hong Kong

Sep 2024 – May 2025

General administration, content, and documentation

- Applied data mining to extract and analyse Chinese insurance data from 200+ PDFs, saving 200+ hours.
- Improved IIQE exam answer prediction via prompt engineering (accuracy 40% → 92%).
- Delivered a food product showcase site with Next.js, Tailwind CSS, and React; sales increased over 100%.
- Authored interactive PowerPoint materials for online design courses; course enrolment up 80%.
- Produced AML web-application documentation following team standards and consistent formatting.

Software engineering and cloud

- Optimised a broker management web app with reusable React components, Cache-Control headers, and server-side rendering; load times improved over 50%.
- Refactored a B2B SaaS platform with RESTful APIs and Neo4j integration for relationship visualisation; about 30% faster client decision-making.
- Deployed Python database update jobs on AWS (S3, Lambda) using Cypher to refresh Neo4j on a schedule.

Web development

- Built AML checking and PEP automation in JavaScript, React, and Python; reduced Open Sanction API spend by HK\$ 10,000+/month.
- Extended broker portal with reusable components (e.g. Support Ticket history, Admin Dashboard), SSR, and caching; load time reduced over 50%.

Data analysis and AI evaluation

- Reviewed AI literature (e.g. Qwen, BERT, LLaMA, Gemma) and summarised findings in Tableau.
- Applied TF-IDF, PCA, *k*-means, SVM, anomaly detection, and topic modelling to insurance data; saved 500+ hours of manual work.
- Used LLMs to extract structured data from 40+ Chinese KYC/CDD documents; improved turnaround by 3+ days.
- Performed ad-hoc analyses in Python and SQL to answer business questions.

Representative platform outcomes (cross-cutting)

- Graph API layer on Neo4j for B2B BFSI KYC automation; Python scraping for sanctions and PEP updates.
- Python ETL pipelines for recurring reports (50,000+ records/cycle); report prep time –40%, turnaround from 2 days to 2 hours.
- Evaluation frameworks for 10+ AI tools/agents on financial data (accuracy, consistency, latency, reliability).

Yi Shing Group Ltd. (YSG) <i>Machine Learning Intern</i>	Science Park, Hong Kong <i>Jun 2024 – Aug 2024</i>
• Real-time fire and smoke detection with YOLOv8 (about 90% accuracy); about 95% after optimisation and migration toward Google Cloud Vertex AI. • Full-stack Flask application connected to Vertex AI; CI/CD for deployments. • Webcam-based live inference; Huawei Cloud ModelArts/OBS pipeline for experiments and storage. • Collaborated with electrical engineering on camera integration and field tests; defined endpoints for load and performance testing.	

Yi Shing Group Ltd. (YSG) <i>Machine Learning Intern</i>	Science Park, Hong Kong <i>Jun 2023 – Aug 2023</i>
• Computer vision models with PyTorch and YOLO on Google Colab; datasets from Roboflow and Kaggle. • Preprocessing, resizing, normalisation; bounding-box annotation; accuracy improved from 50% to 70%. • Raspberry Pi prototype linking detections to a fire curtain; Streamlit app for annotated outputs.	

City University of Hong Kong <i>Student Helper and Mentor</i>	Kowloon Tong, Hong Kong <i>Feb 2023 – Mar 2024</i>
• Exploratory analysis on 20+ ESG reports (trends, risks, opportunities). • Built hkmrs.org with WordPress and CSS; SEO and UI/UX focus. • Delivered a TensorFlow model development workshop for 30+ students and faculty (TED CityU). • Supported ad-hoc tasks for the Talent and Education Department.	

Fori Inc. <i>Technology Resource</i>	Islamabad, Pakistan <i>May 2021 – Jun 2021</i>
• Prototyped real-time traffic pattern analysis with Selenium and BeautifulSoup (Google Maps screenshots) and a HTML/CSS/JavaScript dashboard.	

Projects

Blockchain Web3 E-Wallet (JavaScript, TypeScript, Next.js)	<i>Feb 2025 – Mar 2025</i>
• Web3 app for wallet creation, connecting existing wallets, and secure transfers.	
AI Image Classification (Python, PyTorch)	<i>Oct 2024 – Nov 2024</i>
• ResNet18 and Vision Transformer; 98% intra-class and 94% inter-class accuracy.	
Movie Search and Recommender (Python, JavaScript, jQuery, Flask)	<i>Jun 2024 – Jul 2024</i>
• Flask and Pandas pipelines for preprocessing; REST APIs with genre/cast/year filters and pagination.	
Facial Sentiment Classification (Python, TensorFlow, Keras, OpenCV)	<i>Feb 2024 – Mar 2024</i>
• CNN for happy vs. sad; normalisation and augmentation.	
Nutrition Tracker App (Java, CLI, BlueJ, Visual Paradigm)	<i>Feb 2024 – Mar 2024</i>
• UML (class and use-case diagrams); Factory, Builder, State, Observer; SOLID-oriented CLI application.	
Base kernel project (C, Assembly)	<i>Feb 2024 – Mar 2024</i>
• Priority scheduling and named pipes in a teaching OS kernel; about 30% improvement on benchmark tasks and reliable IPC.	

Skills & Interests

- **Languages:** Python, Go, Rust, SQL, JavaScript, TypeScript, Java, C, C++, Bash, PowerShell, HTML, CSS, VBA, Cypher, Assembly, UNIX.
- **Libraries & frameworks:** Snowflake, React, Redis, Node.js, Next.js, Vue.js, FastAPI, Pydantic, Flask, Selenium, NumPy, Pandas, SciPy, scikit-learn, Spark, Seaborn, Matplotlib, PyTorch, OpenCV, TensorFlow, ONNX Runtime, Ollama, Hugging Face, Transformers, jQuery, Tailwind CSS, Express.
- **Data & AI:** Neo4j, RAG, LLMs, MLOps, LLMops, GenAI, TF-IDF, PCA, clustering, classification, anomaly detection, topic modelling, Tableau, Power BI.
- **Cloud & DevOps:** AWS (S3, Lambda, Redshift, QuickSight), GCP, Azure, Vercel, Docker, Kubernetes, Grafana, CI/CD, Ansible, Linux.
- **Tools:** Git, GitHub, Jira, Confluence, Postman, PostgreSQL, MySQL, MongoDB, ServiceNow, Microsoft 365, Chinese word processing, MCP servers.
- **Practices:** Agile/Scrum, REST APIs, system design, software architecture, technical writing, L3 support, UI/UX, SEO (Word-Press).
- **Languages (spoken):** English (fluent); Cantonese and Mandarin (working proficiency); Urdu and Hindi.

Awards, Leadership & Certifications

- International Student Top Scholarship (HK\$720,000 total), City University of Hong Kong (Jun 2022 – Jun 2026).
- President, Pakistan Students' Association, CityU (Dec 2023 – May 2024).
- Managing Director, Student Committee, Jockey Club Humanity Hall (Hall 1) SRO, CityU (Aug 2023 – Aug 2024).
- CompTIA Security+; Google Cybersecurity Professional Certificate (in progress, Jan 2025 – present).